Sheaf-Theoretic Methods in Groundwater Hydrogeochemistry: A Unified Mathematical Framework for Network Topology Inference, Geochemical Inverse Modeling, and Isotope Source Apportionment

Dickson Abdul-Wahab

February 2, 2026

Disclaimer

Definitions & Defaults

Table 1: Core Definitions and Defaults

Item	Default/Description
Ion vector dimension (n)	$n =$ `len(ion_order)` (default 11: Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4)
Required sample columns	sample_id, site_id, all ions in ion_order, EC, TDS, pH
Units	mmol/L for ions (auto-converted from mg/L or meq/L if specified), μS/cm for EC, mg/L for TDS
Optional columns	Coordinates (x,y or latitude,longitude), elevation/z, head, isotopes, nuclear tracers, temporal fields
CLI reference	Run `hydrosheaf --help` for authoritative options

Troubleshooting (Common Input Issues)

- **Missing required columns:** Ensure sample_id, site_id, all ions, EC, TDS, and pH are present and named exactly.
- **Wrong units:** Confirm ion concentrations are in mmol/L, or specify units for auto-conversion. EC must be in μS/cm, TDS in mg/L.
- **Mismatched site/sample IDs:** Each sample_id MUST be unique; site_id may repeat for multiple samples per site.

- **Missing EC/TDS/pH:** These fields are required for CLI validation and must be present.
- **Ions not matching ion_order:** The ion columns MUST match the order and names in ion_order; check config and input file for consistency.

Abstract

We present a comprehensive mathematical framework for integrated hydrogeochemical analysis in groundwater systems using sheaf-theoretic principles. The framework provides a unified computational approach for: (1) inferring flow network topology and connectivity from incomplete hydraulic head, topographic, and physics-based priors; (2) solving reaction path inverse problems to identify optimal transport processes and parsimonious geochemical reaction sets along network edges; and (3) performing trace-element forensics through Bayesian isotope mixing models and vadose zone travel-time deconvolution. By combining weighted least squares optimization with non-negativity constraints, sparse LASSO regression, thermodynamic constraints from PHREEQC, and Richards equation simulation, the method produces physically consistent, parsimonious models suitable for both reactive transport characterization and contamination source apportionment. We establish theoretical results on the optimality of transport solutions and the convergence of the underlying coordinate descent algorithms. Similar to other open-source projects, the implementation is available at `https://github.com/dabdul-wahab1988/Hydrosheaf`.

Contents

List of Figures

List of Tables

List of Main Variables

Symbol	Description
n	Number of ions in the canonical vector (default 11: Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4)
$\mathbf{x}_v$	Ion concentration vector at node v (mmol/L)
$\mathbf{z}$	Sparse reaction vector (extent of mineral reactions)
S	Stoichiometry matrix (ions $\times$ reactions)
$\mathbf{W}$	Weight matrix (typically diagonal, prioritizing conservative tracers)
γ	Evaporation factor (scaling)
f	Mixing fraction (0–1)
λ	LASSO regularization parameter
ℓ, u	Lower/upper bounds for reaction extents (from SI)
J_{uv}^{sheaf}	Sheaf-consistency score for edge (u, v)
β	Softmax sharpness parameter (`sheaf_soft_beta`)
`sheaf_max_iter`	Max outer iterations for refinement
`sheaf_age_enabled`	Enable age coherence in refinement
`sheaf_weight_age`	Weight for age cost in edge scoring
`sheaf_weight_global`	Weight applied to global residual term in edge scoring
`sheaf_weight`	Soft selection weight stored on each retained edge (edge attribute)

Note: The canonical ion vector is $n = 11$ (Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4), matching the default `ion_order` in code. Some routines (e.g., nitrate source discrimination, unit conversion) treat K specially; see schema and conversion notes below.

1 Introduction

1.1 Overview

Groundwater hydrogeochemistry seeks to understand the chemical evolution of water as it flows through aquifer systems. The inverse problem—determining which combination of transport processes (mixing, evaporation) and geochemical reactions (mineral dissolution/precipitation, redox reactions, ion exchange) explains observed chemical changes—is fundamental to aquifer characterization, contamination assessment, and water resource management.

Consider a groundwater flow network represented as a directed graph $G = (V, E)$, where vertices V represent sampling locations (wells) and edges E represent inferred flow connections. At each vertex $v \in V$, we observe a vector of ion concentrations $\mathbf{x}_v \in \mathbb{R}^n$ (default $n = 11$ columns: Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4 in mmol/L; $n =$ `len(ion_order)` by default, and K is included in `ion_order`).

For each directed edge $(u, v) \in E$, we seek to determine:

1. A transport hypothesis (evaporation scaling γ or mixing fraction f, plus an end-member identifier if mixing is used).

2. A sparse reaction explanation $\mathbf{z}$ consistent with thermodynamic constraints.

To obtain meaningful, physically plausible solutions, we employ three complementary strategies:

- **Sparsity regularization**: Prefer explanations involving fewer reactions (Occam's razor). The L1 penalty in LASSO regression automatically selects a sparse subset of reactions from the candidate dictionary.

- **Thermodynamic constraints**: Enforce mineral equilibrium bounds via saturation indices computed by PHREEQC. This eliminates geochemically impossible solutions.

- **Multi-physics integration**: Incorporate isotope data (δ^{18}O, δ^2H) and electrical conductivity to discriminate between processes.

1.2 Comparison with Existing Tools

The framework presented in this document differs from existing tools in several important ways:

- **Global Parsimony (Occam's Razor)**: Unlike PHREEQC's combinatorial approach which yields all mathematically possible models (often hundreds), Hydrosheaf uses L1 regularization (LASSO) to automatically select the simplest, most physically plausible reaction set.

- **Automated Topology**: Existing codes require the user to explicitly define "Flow Path A to B." Hydrosheaf infers the flow network structure probabilistically from hydraulic heads and topography, making it suitable for regional-scale studies where flow paths are uncertain.

- **Dual-Tier Source Apportionment**: Hydrosheaf is uniquely designed for contaminant forensics, integrating a specialized Bayesian engine for nitrate source discrimination that seamlessly blends hydrochemistry with isotope data.

Table 3: Comparison of Inverse Modeling Capabilities and Features

Feature	**Hydrosheaf**	**PHREEQC**	**NETPATH**	**GWB**
Algorithm	*Sparse L1 Optimization*	Combinatorial Search	Mass Balance	Mass Balance
Network Inference	*Probabilistic Graph*	User-Defined 1D Path	User-Defined	User-Defined
Transport Logic	*Hypothesis Comp. (AIC)*	Fixed Assumption	Fixed Mixing	Fixed
Isotopes	*Dual-Isotope Mixing*	Basic Fractionation	Rayleigh	Basic
Uncertainty	*Bayesian MCMC / Bootstrap*	Sensitivity Analysis	None	Monte Carlo (React)
Nitrate Source	*CoDA + Bayesian Gating*	None	None	None

1.3 System Architecture: Sheaf-Theoretic Perspective

The sheaf-theoretic viewpoint interprets the groundwater network as a cellular complex where local data (ion concentrations at wells) must satisfy global consistency conditions (mass balance, charge balance, thermodynamic equilibrium). While we do not develop full sheaf cohomology here, the computational framework enforces consistency through residual minimization along edges, analogous to minimizing the discrepancy in sheaf-theoretic data structures.

1.4 Input-Output Contract

Input:

- **Samples CSV (CLI)**: Must contain `sample_id`, `site_id`, all ions in `ion_order`, and bulk fields `EC`, `TDS`, `pH`. Optional columns unlock features: coordinates (`x,y` or `latitude,longitude`) for automatic edge inference; head/topo keys for head priors; isotopes for sheaf refinement; conservative tracers (e.g., `Cl`); nuclear tracers (`3H`, etc.) for age coherence.

- Ion concentration vectors: default $n = 11$ columns (Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4), matching `ion_order` in code.

- Optional physics priors from mesh-based flow/particle tracking (FloPy/MODPATH).

Units and Conversion:

- Accepted input units: `mmol/L`, `mg/L`, `meq/L` (if exposed in CLI/config). The CLI supports automatic conversion to mmol/L, including K even if not in `ion_order`.

Output:

- For each edge $(u, v) \in E$: the optimal transport model (evaporation factor γ or mixing fraction f).
- For each edge: sparse reaction vector $\mathbf{z}$, reaction set, and diagnostics (LASSO solution, bounds, SI, etc.).
- For each node: inferred head, isotope, and age estimates (if enabled).

Samples CSV Schema

Table 4: Required Columns (MUST be present)

Column Name	Type	Units	Description
sample_id	string	–	Unique sample identifier
site_id	string	–	Site/well identifier (may repeat for multiple samples per site)
Ions in `ion_order`	float	mmol/L (or convertible)	Ion concentrations; columns MUST match the configured `ion_order` (default: Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4)
EC	float	μS/cm	Electrical conductivity
TDS	float	mg/L	Total dissolved solids
pH	float	–	pH value

Table 5: Optional Columns (unlock features)

Column Name(s)	Feature Unlocked
x, y | latitude, longitude	Automatic edge inference (spatial)
elevation | z	Head/topo priors
head	Hydraulic head prior
δ^{18}O, δ^{2}H, δ^{15}N, δ^{18}O-NO_3	Isotope-based refinement
Cl (if not in ion_order)	Conservative tracer for refinement
3H, tritium, age_years	Age coherence (nuclear tracer)
temporal fields	Time-series analysis (if enabled)

Minimal Valid Example

```
sample_id,site_id,Ca,Mg,Na,K,HCO3,Cl,SO4,NO3,F,Fe,PO4,EC,TDS,pH
S1,WellA,1.00,0.50,0.80,0.10,2.50,1.20,0.60,0.05,0.01,0.00,0.00,700,
450,7.2
S2,WellB,1.10,0.55,0.82,0.10,2.60,1.30,0.62,0.05,0.01,0.00,0.00,720,
470,7.1
```

Note: All required columns MUST be present and named exactly. Units are converted automatically if specified in config/CLI. Optional columns MAY be included to enable advanced features.

2 Governing Mathematical Framework

2.1 Mathematical objects and notation

The main body defines the minimum mathematical objects needed to make the implementation unambiguous and reproducible.

- **Graph:** The groundwater network is a directed graph $G = (V, E)$; each edge $e = (u \to v)$ is a candidate flow connection.
- **Node feature vector:** Each node v has an ion vector $\mathbf{x}_v \in \mathbb{R}^n$, where $n =$ `len(ion_order)`. The default is $n = 11$.
- **Stalks:** The sheaf stalk at each node MUST be $\mathcal{F}(v) = \mathbb{R}^n$. A global assignment (section estimate) is $\mathbf{X} = \{\mathbf{x}_v\}_{v \in V}$.
- **Restriction/transport map:** For each directed edge $e = (u \to v)$, the implementation fits an affine map $\rho_e(\mathbf{x}_u) = \alpha_e \mathbf{x}_u + \mathbf{b}_e$ that predicts downstream chemistry from upstream chemistry under the selected transport+reaction explanation.
- **Soft edge weight:** The refinement procedure assigns a nonnegative soft weight w_e per edge, stored in the edge attribute `sheaf_weight` (also mirrored into `edge_confidence`) and used by the directed section solver.

2.2 Graph-Theoretic Foundation

2.2.1 Probabilistic Edge Weights

In groundwater networks, flow directions are inferred from hydraulic head measurements. Given hydraulic heads h_i, h_j with uncertainties σ_i, σ_j at nodes i, j separated by distance d_{ij}, the head difference is $\Delta h_{ij} = h_i - h_j \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2)$. The probability of flow from i to j is:

$$p(i \to j) = \Phi(\mu_{ij}/\sigma_{ij}) \tag{1}$$

2.2.2 Hydraulic Gradient Guard

To avoid spurious connections, we apply a gradient threshold. If $g_{ij} = |\Delta h_{ij}|/d_{ij} < g_{\min}$, we attenuate the edge probability:

$$p(i \to j) \leftarrow 0.5 + (p(i \to j) - 0.5) \cdot \frac{g_{ij}}{g_{\min}} \tag{2}$$

2.2.3 Three-Dimensional Extensions

For 3D systems, the effective hydrological distance d_{3D} uses a vertical anisotropy factor $\alpha_v \approx \sqrt{K_v/K_h}$:

$$d_{3D} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + \frac{(z_1 - z_2)^2}{\alpha_v^2}} \tag{3}$$

Wells with overlapping screened intervals are assigned higher connectivity probabilities via a screen overlap bonus.

2.2.4 Sheaf Consistency and the Laplacian

In the sheaf-theoretic view, we define stalks $\mathcal{F}(v) = \mathbb{R}^n$ at each vertex and interpret each directed edge $e = (u \to v)$ as imposing a compatibility constraint between upstream and downstream node vectors.

In the implementation, each edge $e = (u \to v)$ carries an affine transport/reaction map $\rho_e(\mathbf{x}_u) = \alpha_e \mathbf{x}_u + \mathbf{b}_e$ and a nonnegative soft weight w_e (stored as the edge attribute `sheaf_weight`). Given a candidate global assignment $\mathbf{X} = \{\mathbf{x}_v\}_{v \in V}$, define the edge residual

$$\mathbf{r}_e(\mathbf{X}) := \rho_e(\mathbf{x}_u) - \mathbf{x}_v.$$

The directed section step solves a weighted least-squares problem with optional anchoring and regularization:

$$\mathcal{E}(\mathbf{X}) = \sum_{e=(u \to v) \in E} w_e \, \|\mathbf{r}_e(\mathbf{X})\|_2^2 \; + \sum_{v \in V_{\text{obs}}} \lambda_v \, \|\mathbf{x}_v - \mathbf{x}_v^{\text{obs}}\|_2^2 \; + \epsilon \sum_{v \in V} \|\mathbf{x}_v\|_2^2. \tag{4}$$

When the maps are linear and anchors are fixed, (4) is a quadratic form analogous to a sheaf Laplacian energy. In practice, Hydrosheaf constructs the corresponding normal equations and solves for $\mathbf{X}$, and the resulting residuals $\|\mathbf{r}_e(\mathbf{X})\|$ are fed back into the refinement score (Section 3.3).

2.3 Transport Processes

Remark 2.1 (Conservative Weighting). The weight matrix $\mathbf{W}$ refers to the conservative weighting configuration ($\mathbf{W}_{\text{cons}}$), typically prioritizing non-reactive tracers (Cl, Br, isotopes).

2.3.1 Evaporation Model

The evaporation model assumes conservative scaling of all solute concentrations:

$$\mathbf{x}' = \gamma \mathbf{x}_u, \quad \gamma \geq 1 \tag{5}$$

Theorem 2.2 (Optimal Evaporation Factor). *For the weighted least squares problem* $\min_{\gamma \geq 1} \|\mathbf{x}_v - \gamma \mathbf{x}_u\|_{\mathbf{W}}^2$*, the optimal solution is:*

$$\gamma^* = \max \left\{ 1, \frac{\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_v}{\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u} \right\} \tag{6}$$

2.3.2 Mixing Model

The mixing model represents dilution or mixing with a distinct water type:

$$\mathbf{x}' = (1 - f)\mathbf{x}_u + f \mathbf{x}_{\text{end}}, \quad f \in [0, 1] \tag{7}$$

Theorem 2.3 (Optimal Mixing Fraction)**.** *For the weighted least squares problem*

$$\min_{f\in[0,1]} \|\mathbf{x}_v - (1-f)\mathbf{x}_u - f\mathbf{x}_{end}\|^2_{\mathbf{W}},$$

the optimal solution is:

$$f^* = \max\left\{0, \min\left\{1, \frac{\mathbf{d}^\top \mathbf{W}(\mathbf{x}_v - \mathbf{x}_u)}{\mathbf{d}^\top \mathbf{W}\mathbf{d}}\right\}\right\} \tag{8}$$

where $\mathbf{d} = \mathbf{x}_{end} - \mathbf{x}_u$.

2.4 Geochemical Reactions

2.4.1 Sparse Reaction Formulation (LASSO)

After transport, the residual mass $\mathbf{r} = \mathbf{x}_v - A(\boldsymbol{\theta}^*)\mathbf{x}_u - \mathbf{b}(\boldsymbol{\theta}^*)$ is explained by mineral reactions. The LASSO problem seeks sparse reaction extents $\mathbf{z} \in \mathbb{R}^m$:

$$\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{r} - S\mathbf{z}\|^2_{\mathbf{W}} + \lambda\|\mathbf{z}\|_1 \quad \text{subject to} \quad \boldsymbol{\ell} \leq \mathbf{z} \leq \mathbf{u} \tag{9}$$

The L1 penalty $\|\mathbf{z}\|_1$ promotes sparsity, yielding a parsimonious explanation involving few reactions.

Derivation of the Coordinate Update To solve (9), we consider the univariate subproblem for z_j, holding all other $z_{k\neq j}$ fixed. Let $\mathbf{r}_j = \mathbf{r} - \sum_{k\neq j} z_k \mathbf{s}_k$ be the partial residual. The subproblem is:

$$\min_{z_j} \frac{1}{2} w_j (r_j - s_j z_j)^2 + \lambda |z_j| \tag{10}$$

Setting the subdifferential to zero, $0 \in -s_j w_j (r_j - s_j z_j) + \lambda \partial |z_j|$, where $\partial|z_j|$ is the signum function if $z_j \neq 0$ and $[-1, 1]$ if $z_j = 0$. Solving for z_j yields the **soft-thresholding operator**:

$$z_j = \frac{\mathcal{S}(s_j w_j r_j, \lambda)}{s_j^2 w_j}, \quad \mathcal{S}(\rho, \lambda) = \text{sign}(\rho)\max(0, |\rho| - \lambda) \tag{11}$$

In the multi-ion case with diagonal weights $\mathbf{W}$, the term $s_j w_j r_j$ generalizes to the dot product $\mathbf{s}_j^\top \mathbf{W}\mathbf{r}_j$.

2.4.2 Thermodynamic Constraints and Stoichiometry

Saturation indices (SI) computed by PHREEQC determine thermodynamically feasible reactions bounds (ℓ_j, u_j):

$$(\ell_j, u_j) = \begin{cases} (0, +\infty) & \text{if } \mathrm{SI}_u < -\tau \text{ and } \mathrm{SI}_v < -\tau \quad \text{(dissolution only)} \\ (-\infty, 0) & \text{if } \mathrm{SI}_v > \tau \quad \text{(precipitation only)} \\ (-\infty, +\infty) & \text{otherwise} \quad \text{(free)} \end{cases} \tag{12}$$

The framework explicitly distinguishes between **Open System** (infinite CO_2 reservoir) and **Closed System** (limited CO_2) carbonate dissolution. For Calcite, the open stoichiometry is $CaCO_3 + CO_2 + H_2O \rightarrow Ca^{2+} + 2HCO_3^-$ (1:2 ratio), while the closed stoichiometry is $CaCO_3 + H^+ \rightarrow Ca^{2+} + HCO_3^-$ (1:1 ratio). Similar variants are provided for Dolomite and Magnesite.

Layer-specific mineral dictionaries are supported for 3D stratified systems.

2.5 Isotope Hydrogeology

2.5.1 Local Meteoric Water Line

Stable water isotopes (δ^{18}O, δ^{2}H) follow the Local Meteoric Water Line (LMWL): $\delta^2\mathrm{H} = a + b \cdot \delta^{18}\mathrm{O}$. Deuterium excess $d = \delta^2\mathrm{H} - 8 \cdot \delta^{18}\mathrm{O}$ quantifies deviation.

2.5.2 Isotope-Based Penalties

For evaporation, we expect $|E_v| > |E_u|$ (deviation from LMWL) and $d_v < d_u$ (decrease in d-excess). Penalties are added to the transport optimization to bias model selection.

2.5.3 Nitrate Source Discrimination

We use two methods for nitrate source attribution:

1. **Dual Isotope Mixing**: Bayesian classification using δ^{15}N and δ^{18}O of nitrate against source endmembers (Manure, Fertilizer, Soil N, Atmospheric).

2. **Hydrochemical Ratios**: When isotopes are unavailable, Compositional Data Analysis (CoDA) and ratios (NO_3/Cl, NO_3/K) are used with Bayesian evidence accumulation.

2.6 Vadose Zone Physics

The optional vadose-zone module simulates 1D unsaturated flow using Richards' equation:

$$C(\psi)\frac{\partial \psi}{\partial t} = \frac{\partial}{\partial z}\left[K(\psi)\left(\frac{\partial \psi}{\partial z} - 1\right)\right] - S(\psi, z, t), \tag{13}$$

where $C(\psi) = \partial\theta/\partial\psi$ is the specific moisture capacity.

2.6.1 Numerical Discretization

We discretize (13) using an implicit finite difference scheme on a cell-centered grid with spacing Δz. Using the backward Euler approximation for time:

$$C_i^{n+1,m}\frac{\psi_i^{n+1,m+1} - \psi_i^n}{\Delta t} = \frac{1}{\Delta z}\left[q_{i-1/2}^{n+1,m} - q_{i+1/2}^{n+1,m}\right] - S_i \tag{14}$$

where m is the Picard iteration index. The inter-node Darcy fluxes q are computed using harmonic means for hydraulic conductivity K:

$$q_{i+1/2} = -K_{i+1/2}\left[\frac{\psi_{i+1} - \psi_i}{\Delta z} - 1\right], \quad K_{i+1/2} = \frac{2K_iK_{i+1}}{K_i + K_{i+1}} \tag{15}$$

The numerical implementation employs a modified Picard iteration with robust boundary condition handling. For Dirichlet (constant head) bottom boundaries, the Darcy flux is computed as $q = K(1 - \frac{\psi_{bc} - \psi_{prev}}{\Delta z})$, while free drainage assumes a unit gradient ($q = K$).

Outputs include vadose-to-groundwater travel-time priors and nitrate breakthrough curves via dual-domain TTD convolution.

2.6.2 Quasi-2D Extensions (Tensor Anisotropy and Seepage Face)

To approximate flow in dipping stratigraphy without a full 2D mesh, Hydrosheaf implements a tensor projection for vertical conductivity. For a layer with dip angle α and anisotropy ratio $A = K_{\parallel}/K_{\perp}$:

$$K_{zz}(\psi) = K_{\perp}(\psi)\left(A\sin^2\alpha + \cos^2\alpha\right) \tag{16}$$

This effectively scales the saturated conductivity K_s to represent the vertical component of the hydraulic tensor, allowing 1D columns to simulate infiltration in sloping aquifers.

Additionally, a dynamic **Seepage Face** boundary condition is implemented to model discharge to surface drains. The boundary switches between a Neumann condition ($q = 0$) when unsaturated ($h < 0$) and a Dirichlet condition ($h = 0$) when saturated, iteratively updating the status based on computed flux direction within the Picard loop.

2.6.3 Transverse Dispersion via Lateral Topology

Standard graph models restrict transport to 1D advection along edges, ignoring transverse dispersion ($D_T\nabla^2 C$) which spreads mass perpendicular to the flow direction. To address this without a mesh, Hydrosheaf employs a **Lateral Topology** inference step.

Neighbors that are spatially close ($d < r$) but hydraulically equipotential ($\nabla h \approx 0$) are classified as "Lateral Edges". These edges do not carry advective flux but serve as potential mixing sources. For a primary flow path $u \to v$, the solver evaluates mixing models where the downstream composition x_v is a mixture of the upstream water x_u and any lateral neighbors x_{lat}:

$$x_v = (1 - f)x_u + f x_{lat} + \Delta x_{rxn} \tag{17}$$

This effectively discretizes the transverse dispersion term into a finite-volume mixing interaction, ensuring that dispersion is only inferred when supported by observed lateral gradients.

3 Numerical Solution Strategy

3.1 Network Optimization Algorithm

The core algorithm employs a nested evaluation strategy. For each edge and each transport model candidate $c \in \mathcal{C}$:

1. **Transport Optimization**: Compute $\boldsymbol{\theta}_c^*$ using conservative weights.
2. **Reaction Optimization**: Solve the bounded LASSO problem for $\mathbf{z}_c^*$ using full weights.
3. **Scoring**: Compute total objective J_c.

The best model is selected via Boltzmann-weighted probabilities: $p_c \propto \exp(-J_c)$.

3.2 Solution of Reaction Equations

Problem (9) is solved via Coordinate Descent (Algorithm 1). To ensure numerical stability, the algorithm includes an adaptive ridge regularization parameter $\epsilon \approx 10^{-10}$ and performs a multicollinearity check.

Specifically, we estimate the condition number κ of the Gram matrix $G = S^\top \mathbf{W} S$. If $\kappa > 30$, a warning is issued indicating that the reaction set may be ill-conditioned, likely due to linear dependence between mineral stoichiometries (e.g., competing silicate weathering proxies).

Algorithm 1 Coordinate Descent for Bounded LASSO

Input: Residual $\mathbf{r}$, stoichiometric matrix S, weights $\mathbf{W}$, penalty λ, bounds $\boldsymbol{\ell}, \mathbf{u}$
repeat
  for $j = 1$ to m **do**
    Compute partial residual ρ_j and normalization ν_j
    Update $z_j \leftarrow \max\{\ell_j, \min\{\mathcal{S}(\rho_j, \lambda/2)/\nu_j, u_j\}\}$
  end for
until convergence

Theorem 3.1 (Convergence of Coordinate Descent)**.** *For the strictly convex bounded LASSO problem in* (9)*, the coordinate descent algorithm (Algorithm 1) converges to the unique global minimizer.*

Proof. The objective function is strictly convex due to the positive definite weight matrix $\mathbf{W}$, and the feasible set is compact. Each coordinate update decreases the objective (or leaves it unchanged), ensuring monotonic convergence. Global convergence follows from standard results on block coordinate descent for convex problems. □

Stability-based tuning selects λ via resampling.

3.3 Graph Inference Algorithms

3.3.1 Bayesian Head Estimation

When direct head measurements are missing, a Bayesian linear-Gaussian model estimates hydraulic heads from topography and depth-to-water priors, propagating uncertainty to edge probabilities.

3.3.2 Sheaf-Consistent Edge Refinement

A sheaf-consistency score J_e^{sheaf} blends local priors and (optionally) a global mismatch signal from the current directed-section solution. For an edge $e = (u \to v)$, the implementation uses the weighted sum

$$J_e^{\text{sheaf}} = w_{\text{head}}\, C_{\text{head}}(e) + w_{\text{iso}}\, C_{\text{iso}}(e) + w_{\text{Cl}}\, C_{\text{Cl}}(e) + w_{\text{age}}\, C_{\text{age}}(e) + w_{\text{global}}\, R_e(\mathbf{X}). \quad (18)$$

Here $C_{\text{head}}, C_{\text{iso}}, C_{\text{Cl}}, C_{\text{age}}$ are edge-local costs (with C_{age} used only when `sheaf_age_enabled` is true), $R_e(\mathbf{X})$ is an edge residual derived from the current directed section (e.g., $\|\rho_e(\mathbf{x}_u) - \mathbf{x}_v\|$), and the scalar weights map directly to configuration parameters:

- `sheaf_weight_head_prior`
- `sheaf_weight_isotope`
- `sheaf_weight_cl`
- `sheaf_weight_age`
- `sheaf_weight_global`

The evaporation probability π_{uv} is refined using three physical heuristics:

1. **Depth Constraint**: Evaporation is attenuated with depth ($z > 30$m).
2. **Thermodynamic Directionality**: Edges inconsistent with SI or redox direction are penalized.
3. **d-Excess Signature**: Kinetic fractionation uniquely decreases d-excess; paths showing d-excess increases are penalized as evaporation candidates.

Candidate edges are scored, and selection uses softmax/Boltzmann weighting:

$$w_e = \frac{\exp(-\beta J_e^{\text{sheaf}})}{\sum_{e'=(u\to\cdot)} \exp(-\beta J_{e'}^{\text{sheaf}})} \quad (19)$$

where β is the sharpness parameter (`sheaf_soft_beta`). The implementation stores w_e into the edge attributes `sheaf_weight` and `edge_confidence`. Very small weights are pruned using a fixed threshold ($w_e > 10^{-3}$), and an outer loop (`sheaf_max_iter`) alternates between rebuilding edge maps and solving a directed section to reduce the global energy (4).

3.3.3 Assumptions and limitations

- **Affine per-edge map:** Each edge is summarized by an affine map $\rho_e(\mathbf{x}_u) = \alpha_e \mathbf{x}_u + \mathbf{b}_e$, which is expressive but not a full reactive-transport simulator.
- **Heuristic costs:** The local cost terms are engineered diagnostics (priors, isotopes, conservative tracers, age) rather than a single generative likelihood.
- **Pruning threshold:** The refinement prunes very small softmax weights using a fixed cutoff (10^{-3}), not a tunable parameter.

Table 6: Sheaf Refinement Parameters

Parameter	Type	Description
sheaf_soft_beta	float	Softmax/Boltzmann sharpness (higher = more selective)
sheaf_max_iter	int	Max outer iterations for refinement loop
sheaf_weight_head_prior	float	Weight for head/topography prior cost C_{head}
sheaf_weight_isotope	float	Weight for isotope compatibility cost C_{iso}
sheaf_weight_cl	float	Weight for conservative-tracer (Cl) cost C_{Cl}
sheaf_age_enabled	bool	Enable age coherence term C_{age}
sheaf_weight_age	float	Weight for age coherence cost C_{age} (when enabled)
sheaf_weight_global	float	Weight for global residual feedback term $R_e(\mathbf{X})$

Algorithm 2 Sheaf-consistent edge refinement outer loop

Input: Candidate directed edges E, observations $\{\mathbf{x}_v^{\text{obs}}\}$, config parameters (Table 6)
Initialize edge weights w_e from priors (e.g., head-based probabilities)
for $t = 1$ to `sheaf_max_iter` **do**
    Fit/update affine edge maps $\rho_e(\mathbf{x}_u) = \alpha_e \mathbf{x}_u + \mathbf{b}_e$ for all retained edges
    Solve directed section: $\mathbf{X}^{(t)} \leftarrow \arg\min_{\mathbf{X}} \mathcal{E}(\mathbf{X})$ (Eq. 4)
    Compute edge residuals $R_e(\mathbf{X}^{(t)})$ from $\rho_e(\mathbf{x}_u) - \mathbf{x}_v$
    Compute local costs $C_{\text{head}}, C_{\text{iso}}, C_{\text{Cl}}, C_{\text{age}}$ (if enabled)
    Form total scores J_e^{sheaf} and softmax weights w_e per source node u
    Prune edges with $w_e \leq 10^{-3}$; store retained weights into `sheaf_weight` and `edge_confidence`
end for
Output: Refined edge set with weights, and final section estimate $\mathbf{X}$

- **Data requirements:** Missing or inconsistent required sample fields (ion order, EC/TDS/pH) can invalidate comparisons and will be rejected by validation.
- **Convergence:** The outer loop is an alternating procedure; while each directed section solve is well-posed, global convergence of the combined refine+refit loop is not guaranteed in general.

3.3.4 Age Coherence (optional)

Purpose: Incorporate groundwater age and nuclear tracer information into edge scoring for improved network inference.

How it works:

- If `sheaf_age_enabled` is true, the refinement algorithm adds an `age_cost` term to the sheaf score for each edge.
- Inputs accepted: `age_years` (numeric), or nuclear tracer fields (any of: `3H`, `tritium`, `tritium_TU`, `tritium_mgL`).
- The `age_cost` penalizes edges inconsistent with expected age gradients or tracer values.
- The weight for this term is set by `sheaf_weight_age` (default 2.0).
- Age coherence improves selection in systems with age data or nuclear tracers, and is fully optional.

Config parameters:

- `sheaf_age_enabled` (bool): Enable/disable age coherence.
- `sheaf_weight_age` (float): Weight for age cost in edge scoring.

Accepted nuclear tracer keys:

- `3H`, `tritium`, `tritium_TU`, `tritium_mgL`

4 Inverse Modeling and Parameter Estimation

4.1 Definition of the Objective Function

For each edge (u, v), we minimize:

$$\mathcal{L}(\boldsymbol{\theta}, \mathbf{z}) = \|\mathbf{x}_v - \mathbf{x}_{\text{pred}}\|^2_{\mathbf{W}} + \lambda \|\mathbf{z}\|_1 + \mathcal{P}_{\text{EC/TDS}} + \mathcal{P}_{\text{iso}} + \mathcal{P}_{\text{Gibbs}} + \mathcal{P}_{\text{cons}} \tag{20}$$

subject to bounds $\boldsymbol{\ell} \leq \mathbf{z} \leq \mathbf{u}$ and $\boldsymbol{\theta} \in \Theta$.

4.2 Optimization Algorithms

For system-wide calibration (e.g. recharge, decay rates), Hydrosheaf supports:

- **Native GLM Engine**: Gauss-Levenberg-Marquardt with trust-region reflective approach.
- **PEST++ Integration**: Wrapper for PEST++-GLM and PEST++-IES for highly parameterized inversion.

4.3 Uncertainty Quantification

4.3.1 Bayesian MCMC Formulation

We formulate the inverse problem in a probabilistic framework to capture the joint posterior distribution of transport and reaction parameters. The model is:

$$\mathbf{x}_v = \gamma \mathbf{x}_u + S\mathbf{z} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma) \tag{21}$$

where $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ represents heteroscedastic measurement and model structural error. The joint posterior is:

$$P(\gamma, \mathbf{z}, \boldsymbol{\sigma} | \mathbf{x}_u, \mathbf{x}_v) \propto P(\mathbf{x}_v | \gamma, \mathbf{z}, \boldsymbol{\sigma}) P(\gamma) P(\mathbf{z}) P(\boldsymbol{\sigma}) \tag{22}$$

We employ a **Laplace Prior** for the reaction extents to enforce sparsity, consistent with the L1 penalty in the deterministic LASSO: $P(z_j) = \frac{1}{2b} \exp(-|z_j|/b)$.

The sampler uses the No-U-Turn Sampler (NUTS) via PyMC. Thermodynamic bounds are implemented using `pm.Truncated` distributions rather than hard potential penalties to maintain smooth gradients for the HMC sampler.

4.3.2 Residual Bootstrap

Bias-Corrected Accelerated (BCa) bootstrap resamples residuals to generate confidence intervals. To ensure physical consistency across replicates, the transport model (e.g., specific endmember or evaporation) is fixed to the original best-fit selection during resampling, preventing the mixing of dissimilar parameter distributions.

5 Problem Definition and Program Implementation

5.1 Implementation Map

The implementation is modular. Key mappings:

- Edge Inference: `hydrosheaf/graph/build.py`
- Transport/Reaction Fitting: `hydrosheaf/inference/edge_fit.py`
- Thermodynamics: `hydrosheaf/phreeqc/`
- Nitrate Source: `hydrosheaf/nitrate_source_v2.py`

5.2 Programmatic API

Primary entry points:

```
from hydrosheaf.inference.network_fit import infer_edges, fit_network
edges = infer_edges(samples, method="probabilistic_map", config=config)
results = fit_network(samples, edges, config=config)
```

5.3 Computational Complexity

Total complexity is $O(|E| \cdot (|\mathcal{C}| \cdot n + T \cdot m \cdot n))$. For a medium network (120 edges), runtime is ≈ 15 min on standard hardware.

5.4 Extension Capabilities

- **Reactive Transport**: Kinetic validation via Damköhler number check.
- **3D Networks**: Layer-aware mineralogy and anisotropic distance.
- **Physics Priors**: Integration with MODPATH endpoints.

5.5 Temporal Deconvolution

Time-varying signals are resolved via:

- **Cross-Correlation**: Estimates mean residence time τ^* from tracer time series.
- **TTD Convolution**: Models downstream concentration as

$$C_v(t) \approx \int C_u(t-\tau) g(\tau) e^{-k\tau} d\tau,$$

  solving for the kernel $g(\tau)$ via nonnegative least squares. For the Piston Flow Model (PFM), where $g(\tau) = \delta(\tau - \tau^*)$, the implementation includes physical validity checks to warn for negative residence times or extrapolation beyond the available input history.

6 Example Problems and Validation

6.1 Validation Methodology

Validation criteria include: Charge Balance Error (¡5%), EC/TDS consistency, Cross-validation (R^2), Thermodynamic consistency, and Isotope coherence.

6.2 Representative Case Studies

6.2.1 Salinization in Irrigated Aquifers

Differentiation between evaporation ($\gamma > 1$, isotope enrichment) and halite dissolution (stoichiometric Na/Cl release).

6.2.2 Nitrate Contamination and Denitrification

Identification of denitrification via negative NO_3 extent coupled with HCO_3 production.

6.3 Benchmarks

Performance scales linearly with edge count. LASSO solver dominates cost but is efficient.

7 Input Data Reference

7.1 Command Line Interface

For authoritative and up-to-date CLI options, run:

```
hydrosheaf --help
```

Typical usage:

```
hydrosheaf --input samples.csv --output results.csv --config config.yaml
```

7.2 Data Schemas

- **Samples CSV**: Must contain `sample_id`, `site_id`, the configured ion columns (default 11-ion set), and `EC`, `TDS`, `pH`. Optional: coordinates, elevation, isotopes, nuclear tracers, time fields.
- **Temporal CSV**: Long-form with `site_id`, `timestamp`, `parameter`, `value`.

8 Output Data Reference

8.1 Output Files

- JSON or CSV format.
- One row per directed edge ($u \to v$).

8.2 Result Tables Structure

Columns include:

- `edge_id`, `u`, `v`: Topology.
- `transport_model`, `gamma`, `f`: Transport parameters.
- `reaction_*`: Reaction extents (mmol/L).
- `chemistry_r2`, `objective_score`: Fit metrics.
- `qc_flags`, `phreeqc_ok`: Diagnostics.

A Mathematical Notes (Optional)

- **Quadratic structure:** For fixed edge maps and weights, the energy in Eq. 4 is a convex quadratic in $\mathbf{X}$ and can be solved via normal equations.
- **Per-ion separability:** With diagonal weights and an affine map that acts identically across ion dimensions (scalar α_e), the solve decouples by ion and can be implemented as repeated solves of the same weighted graph system.
- **Softmax behavior:** The softmax weights $w_e \propto \exp(-\beta J_e)$ interpolate between uniform selection ($\beta \to 0$) and near-hard selection of the lowest-score edge(s) ($\beta \to \infty$).

Final Consistency Checklist

- **Ion vector:** Confirm $n = 11$ and ion_order matches (Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4).
- **Sample CSV schema:** Required columns: sample_id, site_id, all ions, EC, TDS, pH. Optional: coordinates, elevation, isotopes, nuclear tracers, time fields.
- **Units:** Ions in mmol/L (or convertible), EC in μS/cm, TDS in mg/L.
- **Config defaults:** Use default ion_order unless overridden; rely on CLI `--help` for current flags.
- **Refinement mechanics:** Edge refinement uses softmax/Boltzmann selection with `sheaf_soft_beta`; age coherence is optional.
- **Validation:** Run CLI validation to catch schema/units issues before analysis.

B CLI Reference (Auto-Generated)

This section should be generated from the output of
`hydrosheaf --help`
at release time.

C Web-Based User Interface

To make Hydrosheaf accessible to a broader audience beyond command-line users, we have developed a modern full-stack web application (version 0.4.0). This interface provides interactive visualization, project management, and real-time analysis capabilities while maintaining the mathematical rigor of the underlying computational framework.

C.1 Architecture Overview

The web application follows a three-tier architecture:

- **Frontend Layer**: React-based single-page application (SPA) with modern UI components, state management via React Context, and real-time updates via WebSockets.
- **Backend Layer**: FastAPI (Python 3.8+) REST API providing endpoints for project management, sample data validation, edge inference, and asynchronous analysis execution.
- **Computational Core**: The same mathematical framework presented in Sections 2–5.4, ensuring consistency between CLI and web interfaces. All transport optimization, LASSO solving, and thermodynamic constraints use identical implementations.

C.2 Frontend Features

The React-based frontend implements a "Dark Ocean" design theme with glassmorphism effects, providing an intuitive user experience for complex hydrogeochemical analysis:

C.2.1 Interactive Dashboard

The main dashboard provides:

- **Project Management**: Create, load, and manage multiple groundwater analysis projects with persistent browser storage (IndexedDB) or server-side PostgreSQL backend.
- **Quick Statistics**: Real-time summaries of sample counts, inferred edges, and analysis completion status.
- **Recent Activity**: Timeline view of analysis runs, parameter changes, and results exports with version history.

C.2.2 Sample Data Manager

Provides spreadsheet-like interface for:

- **Data Entry**: Input or paste chemical concentrations, coordinates, and metadata with automatic unit conversion (mg/L $\leftrightarrow$ mmol/L, meq/L $\leftrightarrow$ mmol/L).
- **CSV Import/Export**: Bulk upload of field data with schema validation (via Pydantic models) and error highlighting.

- **Quality Control**: Automated charge balance error calculation, outlier detection via robust Z-scores, and completeness validation (missing Ca, Mg, etc.).

C.2.3 Network Visualization

Interactive graph editor using D3.js force-directed layouts:

- **Node Positioning**: Drag-and-drop arrangement with optional geographic coordinate alignment (lon/lat $\rightarrow$ Web Mercator projection).
- **Edge Inference Controls**: Visual configuration of probabilistic edge inference parameters (radius km, minimum head gradient, p_{min} threshold).
- **3D Layer Support**: Toggle vertical stratification view and visualize cross-layer connectivity (requires `--3d` mode enabled).
- **Edge Attributes**: Color-coded edges by confidence probability, dominant transport model (evaporation/mixing), or total reaction extent $\|\mathbf{z}\|_1$.

C.2.4 Analysis Configuration Panel

Formalized interface for model parameters mapped to CLI flags:

- **Transport Models**: Select candidate endmembers (rainfall, seawater, river), enable/disable evaporation hypothesis, configure isotope penalty weights (η_E, η_d).
- **Reaction Dictionary**: Toggle minerals (calcite, gypsum, halite, silicates), set custom stoichiometry, select PHREEQC thermodynamic database (WATEQ4F, PHREEQC.dat, llnl.dat).
- **Regularization**: Adjust L1 penalty (λ), enable residence-time coupling (`--residence-coupling`), set coordinate descent convergence tolerance.
- **Advanced Options**: Enable temporal mode (`--temporal-enabled`), select uncertainty quantification method (Bootstrap/Bayesian MCMC), upload vadose priors CSV.

C.2.5 Real-Time Analysis Monitor

Asynchronous execution with live feedback via WebSockets (FastAPI's `WebSocketEndpoint`):

- **Progress Tracking**: Per-edge status (queued, running, complete, failed) with estimated time remaining based on historical per-edge solve times.
- **Intermediate Results**: Streaming updates of transport model selection, LASSO iteration count, reaction convergence flags, and current objective value.
- **Log Viewer**: Real-time display of solver diagnostics, PHREEQC warnings (saturation index failures, database issues), and QC flags (charge balance errors, isotope inconsistencies).

C.2.6 Results Dashboard

Comprehensive visualization of analysis outputs:

- **Summary Tables**: Sortable, filterable grids (AG Grid React) of edge results with dynamic columns: transport parameters (γ, f, endmember ID), reaction extents (one column per active mineral), fit metrics (R^2, residual norm, L1 norm).
- **Spatial Maps**: Leaflet-based choropleth visualization of transport model prevalence, TDS evolution gradients, nitrate source probabilities (P_{manure}).
- **Piper/Stiff Diagrams**: Automated hydrochemical facies classification via ternary projections and evolution trajectories along flow paths.
- **Time-Series Plots**: When temporal mode is enabled (`--temporal-enabled`), interactive Plotly charts of residence time estimates, seasonal decomposition (trend + periodic + residual), and TTD kernel visualizations (histogram of $g(\tau)$).
- **Uncertainty Bands**: Credible intervals (Bayesian MCMC) or bootstrap percentiles (BCa) overlaid on predicted concentrations when `--uncertainty` is enabled.

C.3 Backend API

The FastAPI backend exposes RESTful endpoints and WebSocket channels designed around resource-oriented patterns:

C.3.1 Project Management Endpoints

```
POST   /api/projects              Create new project
GET    /api/projects/{id}         Retrieve project metadata
PUT    /api/projects/{id}         Update configuration
DELETE /api/projects/{id}         Delete project
GET    /api/projects/{id}/export  Export as PDF report
```

C.3.2 Sample Data Endpoints

```
POST /api/samples            Upload CSV or JSON sample data
GET  /api/samples/validate   Run schema and QC checks
PUT  /api/samples/{id}       Update individual sample
```

C.3.3 Security Architecture

The backend implements several security measures to protect the integrity of scientific data and computational resources:

- **Rate Limiting**: All API endpoints are protected by a token-bucket rate limiter (default 200 requests/minute) using `slowapi` to prevent abuse and denial-of-service attacks.
- **Input Sanitization**: Project identifiers and file paths are rigorously sanitized to prevent directory traversal attacks.

- **Validation Layer**: Pydantic schemas enforce strict type checking and range validation on all inputs before they reach the core computational engine.
- **Error Handling**: Production error handlers suppress stack traces to prevent information leakage while logging detailed diagnostics for administrators.

C.3.4 Network Endpoints

```
POST /api/edges/infer  Trigger probabilistic edge inference
GET  /api/edges         Retrieve inferred edges with p(i→j)
PUT  /api/edges/{id}    Manually override edge (enable/disable)
```

C.3.5 Analysis Endpoints

```
POST /api/analysis/run            Start analysis (returns job_id)
WS   /api/analysis/ws/{job_id}   WebSocket for real-time progress
GET  /api/analysis/status/{job_id}  Poll job status
GET  /api/analysis/results/{job_id} Retrieve completed results
```

C.3.6 Data Transformation Layer

The backend implements automatic conversions to maintain compatibility with the core framework:

- **Unit Conversion**: Frontend accepts mg/L (common in water quality reports), backend converts to mmol/L using molar masses (e.g., $M_{\mathrm{Ca}} = 40.08$ g/mol) before invoking `hydrosheaf.inference.network_fit`.
- **Schema Validation**: Pydantic models enforce required fields (`site_id`, `Ca`, `Mg`, `Na`, `HCO3`, `Cl`, `SO4`, `EC`, `TDS`, `pH`) and reject malformed inputs with HTTP 422 before reaching the solver.
- **Result Serialization**: Converts NumPy arrays (`ndarray`) and NetworkX graphs (`DiGraph`) to JSON-serializable formats (lists of dicts) for frontend consumption.

C.4 Deployment and Scalability

C.4.1 Development Mode

For local testing and demonstration:

```
# Terminal 1 (Backend)
cd web/backend
uvicorn app.main:app --reload --port 8000

# Terminal 2 (Frontend)
cd web/frontend
npm run dev  # Vite dev server on port 5173
```

Open browser to `http://localhost:5173`. Frontend proxies API requests to `http://localhost:8000`.

C.4.2 Production Deployment

Recommended stack for institutional deployment (university research group, government agency):

- **Reverse Proxy**: Nginx or Traefik for HTTPS termination (Let's Encrypt certificates), load balancing, and static asset serving.
- **Application Server**: Gunicorn with Uvicorn workers (4-8 workers for CPU-bound LASSO solving) for concurrent request handling.
- **Task Queue**: Celery + Redis for background LASSO/MCMC jobs exceeding HTTP timeout limits (default 30s for web APIs).
- **Database**: PostgreSQL 12+ for persistent project storage, user sessions, and analysis result caching. SQLite acceptable for single-user deployments.
- **File Storage**: S3-compatible object storage (MinIO, AWS S3) for large result exports (CSV/JSON > 10 MB) and time-series data.

C.4.3 Performance Considerations

For networks with > 100 edges or Bayesian MCMC uncertainty quantification (4000+ samples per edge):

- **Asynchronous Execution**: Long-running analyses (estimated time > 30s) execute in Celery background workers, preventing frontend timeout.
- **Incremental Results**: WebSocket streaming allows users to view partial results (e.g., first 10 edges completed) while remaining edges compute in parallel.
- **Caching**: Redis-backed result cache (TTL 1 hour) prevents redundant computation when users toggle visualization options (Piper vs. Stiff diagrams) without changing model parameters.
- **Database Indexing**: PostgreSQL B-tree indexes on `project_id`, `job_id`, and `edge_id` enable sub-millisecond lookups for result retrieval.

C.5 Integration with Existing Workflows

The web interface is designed to complement, not replace, the CLI:

- **Bidirectional Compatibility**: Projects created via web UI can be exported as CLI-compatible JSON configuration files (`hydrosheaf --config exported_config.json`), and vice versa (CLI results can be imported into web UI for visualization).
- **Programmatic API**: Advanced users can bypass the web frontend and interact directly with the FastAPI backend via Python `requests` library or R `httr` package for automated workflows.
- **Batch Processing**: For large regional studies (hundreds of wells, dozens of time steps, MCMC uncertainty), the CLI remains preferred due to lower overhead and easier integration with HPC schedulers (SLURM, PBS).

C.6 Accessibility and Usability

The web interface follows modern accessibility best practices:

- **WCAG 2.1 Level AA Compliance**: Sufficient color contrast ratios (4.5:1 for text), keyboard navigation support, ARIA labels for screen readers.
- **Responsive Design**: Mobile-friendly layouts (Tailwind CSS breakpoints) for tablet and phone access, though desktop recommended for complex network editing.
- **Internationalization (i18n)**: React-i18next framework supports multiple languages (English, Spanish, French planned for future releases).

C.7 Future Development

Planned enhancements to the web interface (see `web/INTEGRATION_ROADMAP.md` for detailed timeline):

- **Collaborative Features**: Multi-user access with role-based permissions (viewer, analyst, admin) using OAuth2/OIDC authentication (Keycloak, Auth0).
- **Interactive Tutorials**: Guided walkthroughs using synthetic datasets (similar to `hydrosheaf_synthetic_csv/`) to teach inverse modeling concepts interactively.
- **3D Visualization**: WebGL-based rendering (Three.js) of layered aquifer networks and vertical transport pathways, replacing 2D projections.
- **Parameter Sensitivity Analysis**: Interactive dashboards showing how λ, weights $\mathbf{W}$, and thermodynamic bounds affect sparsity $|\mathcal{A}|$ and fit quality R^2.
- **Integration with External Databases**: Connectors for USGS Water Quality Portal API, state monitoring databases (e.g., California GAMA), and remote PHREEQC thermodynamic data repositories.
- **Live Model Comparison**: Side-by-side comparison of multiple analysis runs (e.g., varying λ from 0.01 to 1.0) with synchronized visualizations.

C.8 Open-Source Licensing and Contributions

The web application (both frontend and backend) is released under the same MIT License as the core Hydrosheaf package. Contributions welcome via GitHub pull requests. Frontend uses npm for dependency management (React 18, Vite 4, Tailwind CSS 3); backend uses Poetry for Python dependency management (FastAPI 0.104, Pydantic 2.5).

Acknowledgments

This work synthesizes methods from convex optimization, geochemistry, and hydrogeology.

References

[1] Lasaga, A. C. (1998). *Kinetic Theory in the Earth Sciences.*

[2] Gelman, A., et al. (2013). *Bayesian Data Analysis.*

[3] Efron, B. (1987). Better Bootstrap Confidence Intervals.

[4] Friedman, J., et al. (2007). Pathwise coordinate optimization.

[5] Friedman, J., et al. (2010). Regularization paths for generalized linear models.

[6] Parkhurst, D. L., & Appelo, C. A. J. (2013). PHREEQC version 3.

[7] Plummer, L. N., et al. (1994). NETPATH version 2.0.

[8] Aqueous Solutions LLC. (n.d.). *The Geochemist's Workbench.*

[9] Wang, Y., et al. (2025). Enhancing inverse modeling in groundwater systems.

[10] Gholami, R., et al. (2024). Geochemistry and machine learning.

[11] Harbaugh, A. W. (2005). MODFLOW-2005.

[12] Pollock, D. W. (2016). MODPATH Version 7.

[13] Bakker, M., et al. (2016). Scripting MODFLOW using FloPy.

[14] PyMC Developers. (2023). PyMC.

[15] ArviZ Developers. (2023). ArviZ.

[16] van Genuchten, M. T. (1980). Hydraulic conductivity of unsaturated soils.

[17] Mualem, Y. (1976). Predicting hydraulic conductivity.

[18] Feddes, R. A., et al. (1978). Simulation of Field Water Use.

[19] Richards, L. A. (1931). Capillary conduction.

[20] Robinson, M. (2008). *Sheaves in Geometry and Logic.*

[21] Appelo, C. A. J., & Postma, D. (2005). *Geochemistry, Groundwater and Pollution.*

[22] Clark, I. D., & Fritz, P. (2015). *Environmental Isotopes in Hydrogeology.*

[23] Kendall, C. (1998). Tracing nitrogen sources.

[24] Xue, D., et al. (2009). Application of stable isotopes.

[25] Curry, J. M. (2014). Sheaves, cosheaves and applications.

[26] Tseng, P. (2001). Convergence of coordinate descent.